



Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
Integrated Bachelor's-Master's Program
Mid-Semester Examination-Autumn 2025

Subject: Energetics and Bonding
Full Marks: 25

Subject Code(s): CHS1101
Time Allotted: 2 hrs

Answer all questions
All questions carry equal marks
(All the symbols bear usual their meaning)

1. (a) Establish $\frac{\partial H}{\partial P_T} = V - T \frac{\partial V}{\partial T_P}$, the thermodynamic equation of state from the two laws of thermodynamics.
(b) Calculate $\frac{\partial H}{\partial P_T}$ for a non-ideal gas obeying $P(V - b) = RT$. (4+1)
2. (a) One mole of an ideal gas with specific heat $C_V = 5R/2$ at a temperature 100°C is compressed under isochoric condition to reach 150°C . Calculate q , w , ΔE and ΔH .
(b) Show that for the same change of state two-stage expansion of a gas gives more work than that for the one-stage expansion. (3.5+1.5)
3. (a) Show that the statement "Heat can flow spontaneously from a low temperature region to a high temperature region" contradicts the Second Law.
(b) Show that the Joule-Thomson expansion is iso-enthalpic. (2.5 +2.5)
4. (a) Use the two laws of thermodynamics to prove $\frac{\partial V}{\partial T_P} = -\frac{\partial S}{\partial P_T}$
(b) Calculate the entropy of a binary mixture of 'A' and 'B' relative to its pure components. Under what condition the entropy change is maximum? (2.5 +2.5)
5. Consider the following reversible cycle with an ideal gas. (1+4)
 - (i) Isothermal compression from a volume V_1 to V_2 at T_C ;
 - (ii) Isochoric pressure increase at a constant volume V_2 ;
 - (iii) Isothermal expansion from volume V_2 to V_1 at T_H ;
 - (iv) Isochoric pressure decrease at a constant volume V_1 .Plot $P - V$ diagram and calculate the efficiency.